SENSOR REPORT

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***Abstract*— This report examines the sensors and supporting hardware used in a Formula Student driverless racecar, including IMUs, GNSS, LiDAR, cameras and wheel-speed sensors. It discusses their working principles, applications, limitations, error mitigation and integration into the vehicle. The report also evaluates key components, including the VectorNav VN-200, Hesai Pandar40P, ZED 2i stereo camera and NVIDIA Jetson AGX Orin, along with costs and sourcing options in India.**

I. INTRODUCTION

A sensor is a device that detects or measures a physical quantity and converts it into a signal that can be interpreted. A formula student driverless car relies on sensors to perceive its surroundings, estimate its position and motion, and navigate the track. Sensors can be classified into:

Perception: LiDAR, mono camera, stereo camera

Localization: GNSS, INS/IMU

Vehicle state: wheel speed, steering angle, brake pressure

Computing platform: NVIDIA Jetson AGX Orin runs all the software

II. IMU – INERTIAL MEASUREMENT UNIT

An Inertial Measurement Unit (IMU) is a sensor that measures a vehicle's acceleration and angular velocity using accelerometer and gyroscope. An accelerometer has a microscopic proof mass which is suspended by springs. Acceleration of the car causes relative movement of the mass, changing an electrical property such as capacitance. The electronic circuits convert this change into a measurement of specific force along each axis. A gyroscope has a microscopic structure which vibrates inside the sensor. Rotation of the car produces a Coriolis effect, which is detected and converted into angular velocity. A magnetometer is an optional sensor which helps estimate the heading of the car with the help of earth's magnetic field. It is helpful when GNSS heading is not reliable.

The IMU processes these measurements into digital outputs. Accelerometers measure specific force while gyroscopes measure angular velocity. Combining their measurements helps estimate vehicle orientation and motion.

A. Applications:

The IMU supports vehicle-state estimation by providing acceleration and rotation measurements. These help the software monitor braking, acceleration and cornering, estimate yaw rate, support path tracking, and combine inertial measurements with GNSS, wheel-speed sensors andcamera or LiDAR data

B. Error Sources and Why Fusion Is Needed Errors are scale-factor error, axis misalignment, temperature drift, cross-axis sensitivity and vibration rectification, which is a particular problem on a single-cylinder FS engine.

C. Use Cases onthe Car :

• Vehicle dynamics: lateral and longitudinal acceleration (g-g diagram), yaw rate and roll and pitch for tyre and suspension analysis.
• State estimation: yaw rate and acceleration drive the EKF that estimates velocity and side-slip angle between GNSS fixes or wheel-speed updates.
• Driverless: ego-motion compensation of LiDAR point clouds and dead-reckoning when vision is poor .
• Traction and launch control: acceleration is used to cross-check wheel speed during wheel spin.

III. GPS AND GNSS RECEIVERS

A. Working Principle:

A Global Navigation Satellite System receiver measures the travel time of signals broadcast by satellites. A standalone receiver achieves metre-level accuracy; the ZED-F9P quotes better than 1.500 m in standalone mode
Real-time kinematic (RTK) positioning improves this to centimetres. A fixed base station or an NTRIP network supplies RTCM correction data, the rover resolves the integer number of carrier-phase cycles to each satellite, and the position error falls to about 0.010 m plus 1 ppm of the baseline. On a 10 km baseline this is 0.020 m [9]. Multi-band (L1/L2) receivers resolve ambiguities in seconds and are far more robust than single-band units. Two receivers in a moving-base arrangement also give true heading from the vector between two antennas, which does not depend on motion or magnetometers. Constellations include GPS, GLONASS, Galileo and BeiDou. India operates NavIC (IRNSS), a regional system covering India and its surroundings, and GAGAN, an augmentation service. Most low-cost multi-band modules, including the ZED-F9P, list GPS, GLONASS, Galileo, BeiDou, QZSS and SBAS but not NavIC [9], so NavIC cannot be assumed for such parts.

B. Use Cases onthe Car:

Helps to time the laps, track mapping and making sure no crahses happen and racing-line comparison from logged position.
• Ground-truth speed to calibrate wheel speed sensors and to measure slip.
• Driverless: ground truth for evaluating SLAM, and global-frame velocity for state estimation

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IV. WHEEL SPEED SENSORS

A. Working Principle

A wheel speed sensor generates a pulse train as a toothed ring (tone wheel) or a set of magnets rotates past it. Three technologies are common. A variable-reluctance (VR) or inductive pickup generates an AC voltage whose amplitude and frequency grow with speed; it is simple and rugged but needs signal conditioning and gives a weak signal at low speed. A Hall-effect sensor is active and outputs a clean digital waveform down to zero speed, which is why modern ABS systems use it. Optical or magnetic encoders give much finer angular resolution but are less tolerant of dirt and heat.

Use Cases on the Car :

Vehicle speed estimation for the EKF, and odometry for driverless SLAM.

B. Candidate Parts, Cost and Sourcing

The most economical option for learning is the A3144E Hall switch, listed at ₹59.00 including GST by Probots (₹50 plus GST), and it needs a magnet ring on the wheel. Inductive barrel sensors such as the LJ12A3-4-Z/BX (NPN, 6 to 36 V, 4 mm nominal sensing distance) are listed in India at about ₹159 to ₹412, for example ₹294 including GST for the M12 variant at Probots, but their switching frequency is limited, so their datasheet must be checked against the highest wheel frequency. For the car itself, automotive active ABS-type Hall sensors (from Bosch, Honeywell, Allegro or Melexis ICs on a custom bracket) are recommended. They are not catalogue items on the hobby shops, so they are sourced through automotive spares dealers or from Mouser India, DigiKey India or Element14 India, and a realistic estimate is a few thousand rupees per set of four.

V. CAMERAS: MONOCULAR AND STEREO

A. Working Principle

A camera projects the 3-D world onto a 2-D sensor. Under the pinhole model a point at (X, Y, Z) maps to pixel coordinates Some intrinsic parameters and the lens distortion coefficients are found by calibration, and extrinsic calibration gives the pose of the camera relative to the car [10]. Important sensor properties are resolution, frame rate, dynamic range, field of view and shutter type. A global shutter exposes all pixels simultaneously, whereas a rolling shutter exposes row by row and skews fast-moving scenes; on a car moving at 20 m/s a global shutter is strongly preferred.

B. Monocular Cameras

A single camera gives rich semantic information (cone colour, signs, track edges) but no direct depth. Depth can be inferred when the object size is known

C. Stereo Cameras

Two cameras separated by a baseline B observe the same scene are stereo cameras basically.

D. Use Cases on the Car:

Driverless cone detection, colour classification and, with a LiDAR, sensor fusion of semantics and geometry .

- Visual odometry and SLAM, with an IMU for scale and robustness.
- Driver-facing or on-board video synchronised with logged data for post-run analysis.

VI. LIDAR

A. Working Principe:

LiDAR (Light Detection and Ranging) measures distance by timing a laser pulse. In the time-of-flight (ToF) method the range is $d = c\,\Delta t / 2$ (10) so an object at 20 m returns the pulse after 133.426 ns, and a timing resolution of 1 ns corresponds to a range resolution of 0.150 m. Some sensors use amplitude-modulated continuous wave (phase shift) or frequency-modulated continuous wave (FMCW) ranging, and low-cost 2-D scanners use laser triangulation, in which the position of the returning spot on a sensor line gives the range. A spinning multi-beam unit stacks 16 to 128 laser channels vertically and rotates them to produce a point cloud of hundreds of thousands of points per second, while solid-state and non-repetitive-scan units (such as Livox) trade field of view for lower cost and no moving parts. Each point also carries a reflectivity (intensity) value, which helps to separate cones from the ground and their colour bands.

B. Procesing Pipeline:

A typical cone-detection pipeline first removes the ground plane (RANSAC or grid-based methods), then clusters the remaining points (Euclidean clustering or DBSCAN), filters clusters by size and shape, and finally classifies them, optionally by projecting the clusters into the camera image for colour [6]. Motion distortion is corrected using the IMU, and the LiDAR must be accurately extrinsically calibrated to the camera and the vehicle frame.

C. Use Cases on the car:

- Driverless cone detection and mapping; the primary perception sensor for most top teams .
- SLAM and localisation against the cone map, together with IMU and wheel speed.
- Obstacle detection for safety-critical stopping decisions.

D. Candidate Parts, Cost and Sourcing

Entry-level USB webcams or Raspberry Pi cameras are adequate for developing the software pipeline, but they use rolling shutters. For the car, global-shutter machine-vision cameras (Basler, FLIR, Allied Vision, or global-shutter OV9281-based modules) with a fixed lens are preferred. For stereo, Intel RealSense D435 and D435i cameras (wide field of view, global-shutter depth sensors, 50 mm baseline) are available in India through ThinkRobotics, where the D435/D435i/D455 family is listed in the range of roughly ₹35,000 to ₹60,000 (the D435i at about ₹43,999.99 on sale). Stereolabs ZED 2i (120 mm baseline) and Luxonis OAK-D Lite cameras are imported through distributors or directly; as a rough estimate (to be re-verified) they cost about US$450 to 550 and US$150 to 200 respectively before shipping, GST and customs duty. Cameras also require a compute unit (an NVIDIA Jetson-class module or a laptop-class computer), which is often more expensive than the camera itself.

VII. CONSOLIDATED COST AND SOURCING IN INDIA

The prices below were read from Indian retailer listings during the research for this report (October 2026) and are shown as listed; they change often and must be re-checked before an order. The ZED-F9P receivers by GenX Battery are listed without a currency symbol in the captured text and are assumed to be in INR. ArduSimple prices are in euros and exclude shipping, GST and customs duty.

- GY-521 MPU-6050: ₹199.00 to ₹274.94 including GST (Probots, Zbotic).
- GY-87 10-DOF IMU: ₹949.00 including GST (Probots).

Buying advice for India follows from these listings. Hobby modules are cheap and available within one to three days from Bengaluru, Delhi and Mumbai suppliers such as Probots, Robu, ThinkRobotics, Evelta and Zbotic, but quality varies and clones are common, so buy critical chips from authorised distributors (Mouser India, DigiKey India, Element14 India, RS Components) and request a GST invoice, because it supports the team's expense records. Imported competition grade parts attract basic customs duty and 18 % IGST and take weeks, so orders should be placed early, and manufacturers frequently offer academic discounts or sponsorship to Formula Student teams. A staged purchase strategy keeps the budget under control. Stage 1 (bench and learning) uses a GY-521, a Hall sensor and a low-cost LiDAR or webcam, costing a few thousand rupees in total. Stage 2 (data-acquisition car) adds a BNO055 or a better IMU, a ZED-F9P receiver and the chassis sensors from Section VIII. Stage 3 (driverless) adds a global-shutter or stereo camera, a multi-channel LiDAR and the compute platform, which dominates the total cost.

VIII. CONCLUSION

A driverless car needs sensors at three levels: rules-driven safety sensors, data-acquisition sensors for the feedback and perception sensors for autonomy. IMUs, GNSS receivers and wheel speed sensors form the core of state estimation, while cameras and LiDAR provide perception. In India, low-cost modules are readily available within like some days, whereas RTK receivers are available locally from Indian assemblers, and cameras and LiDARs mostly require imports, early ordering and sponsorship.

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